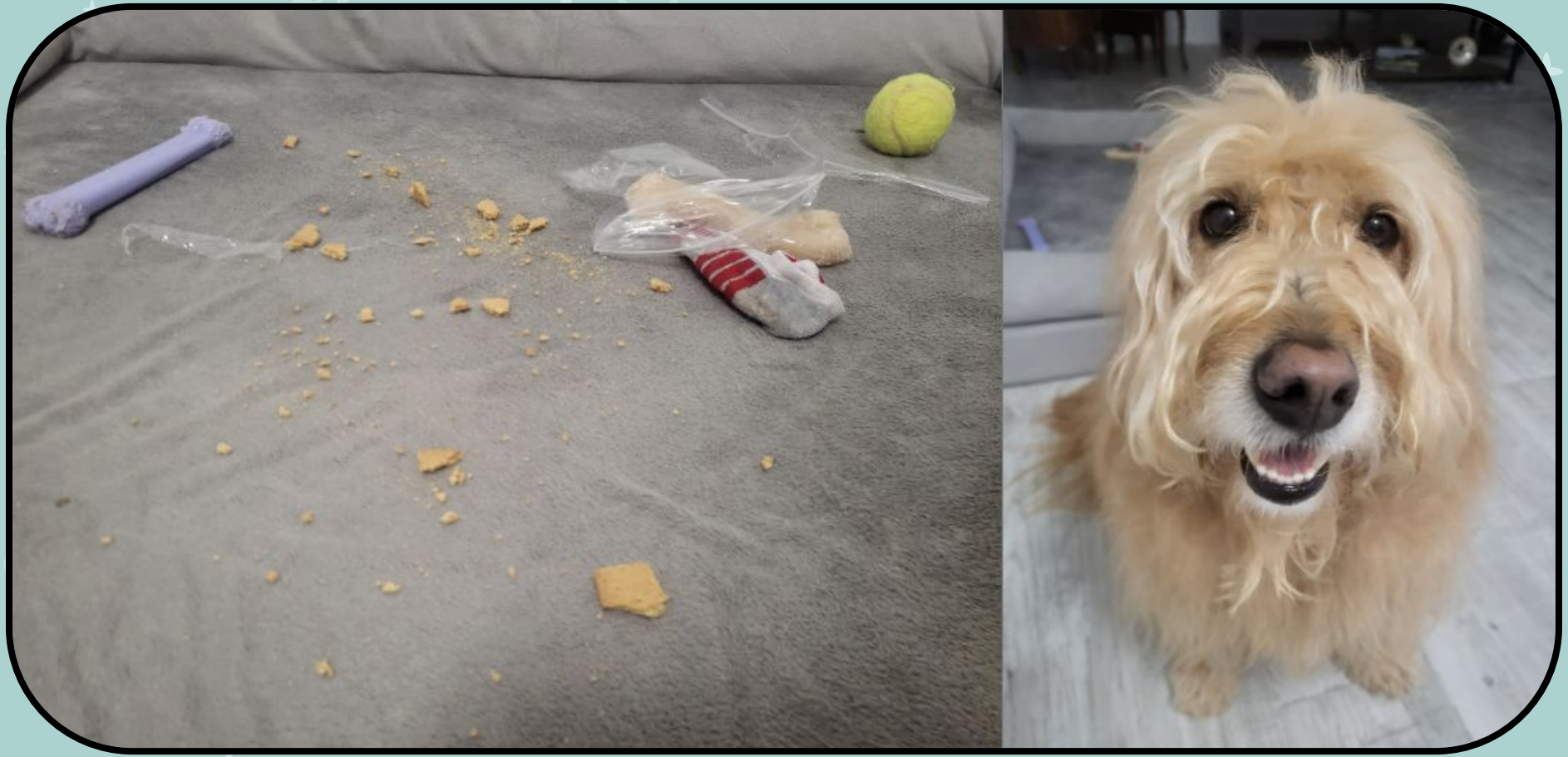
P.A.W.S.

Pet Automated Watch & Security

The Good Boy Gadget







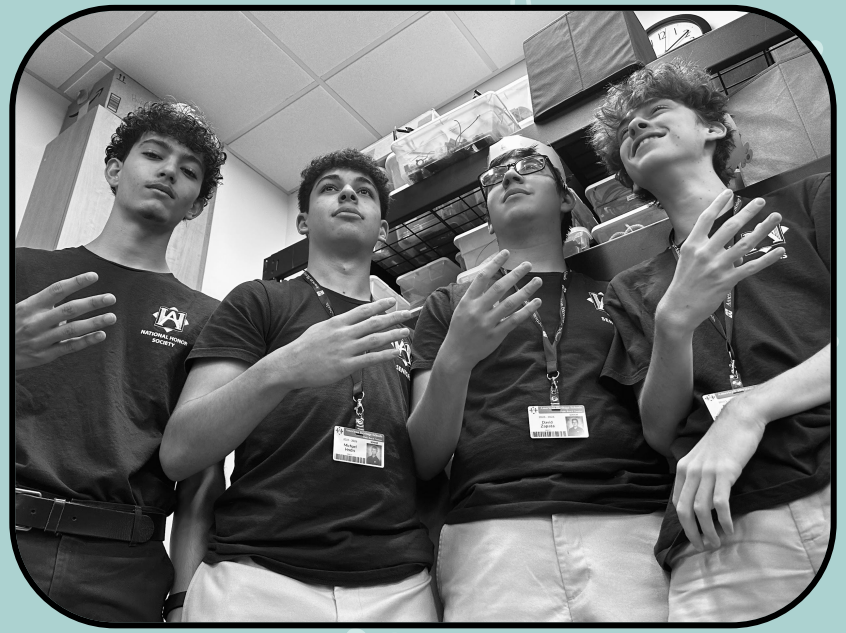
Meet The Team

Michael

Electronics & App Design

Patrick

Electronics & Device Programming



Jacob

CAD/Assembly/ Manufacturing

David

CAD/Assembly/ Manufacturing

Meet The Team

Bailey
Treat Coordinator



Penny
Toy Coordinator



Background

Pets While at Home...

Depression

Inactivity

Separation Anxiety

Malnutrition



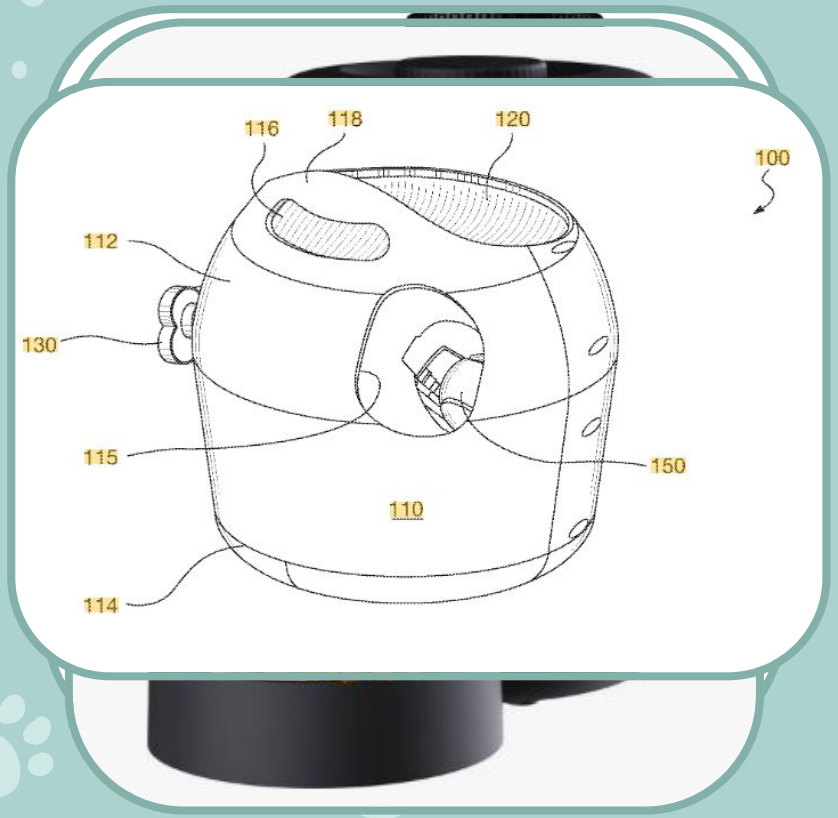
Problem

Over 95% of pets spend their time alone at home inactively resting. This directly affects pet owners, with 75% reporting a negative impact on their pets after unsupervised downtime.





Current Solutions



Ball Launcher

A stationary ball launcher meant to help exercise dogs by intaking from a funnel in the top and launching out the side.

Cons:

- Completely stationary
- Requires human input (so why not just throw it?)
- Bulky

Pet Feeder

A pressed feeder that only releases food dogs and cats.

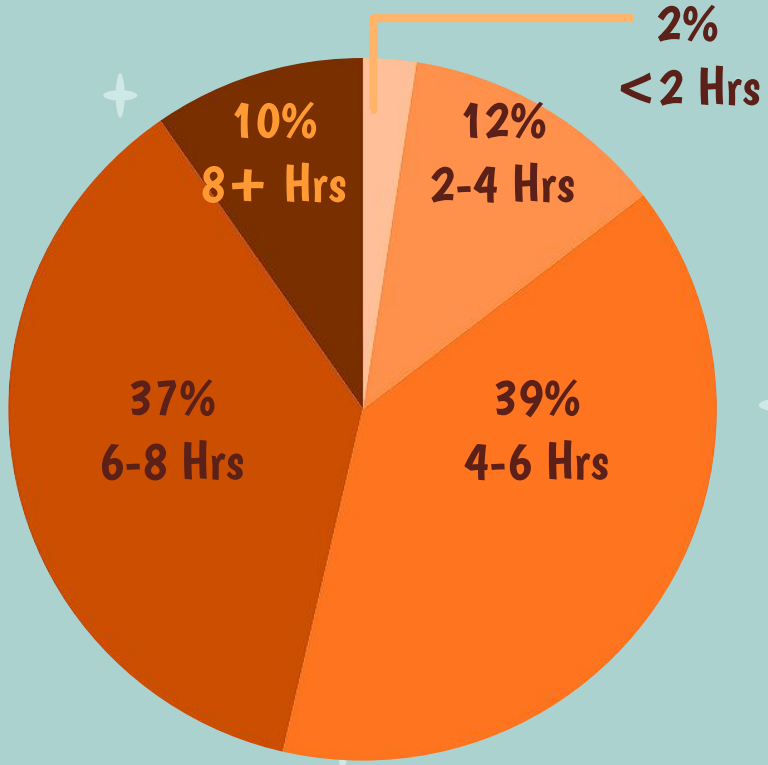
Cons:

- Forced to give the dog pills
- Doesn't work 100% of the time
- May have unwanted side effects
- Based on a set schedule (difficult to change timing)

Hot Rod Dog Calming Anxiety Relief

Pheromone collar that releases calming pheromones to help dogs with anxiety and stress.

Market Research



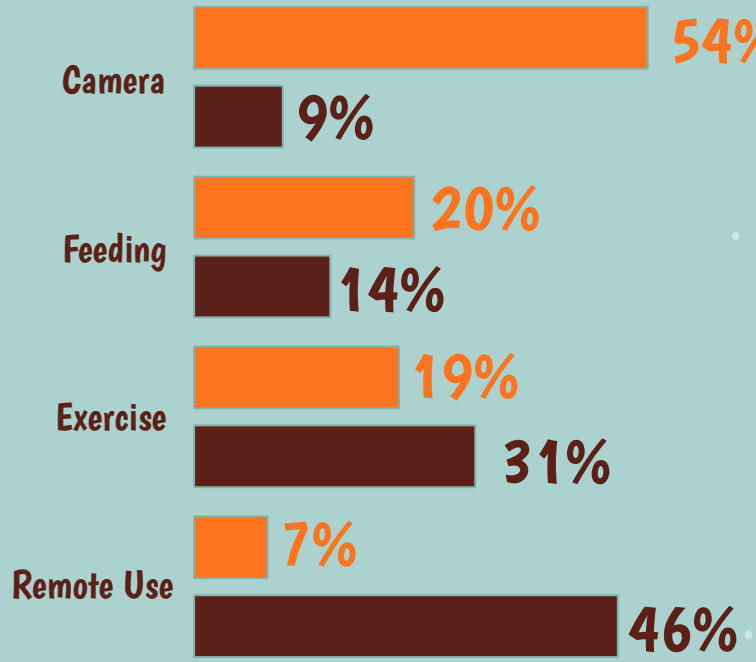
86% Of Dogs are kept home alone for 4 hours or longer on a typical weekday

Market Research

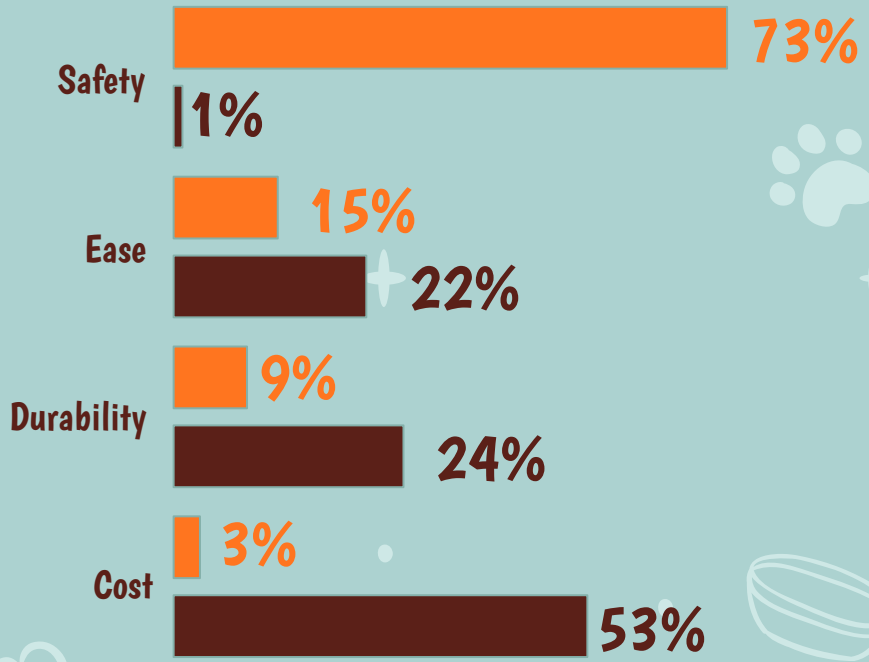
Feature Priority

Most

Least



Product Specification Priority



Expert Interview

Jessica Dreyfuss

- Licensed Veterinarian and manages a Pet Rehabilitation Center
- Add voice recordings to the product to make it seem more alive to the dog
- Add nutritional supplements/oils into feeder mix



Dr. Nashaat Toma

- Retired Veterinarian, Jog Lantana Animal Clinic
- Underlying cause for a majority of pet anxieties: overstimulation of the nervous system
- Environmental safety cueing: reduces hypervigilance/scanning threat
- Suggested light music from the machine for this purpose



Generation 1





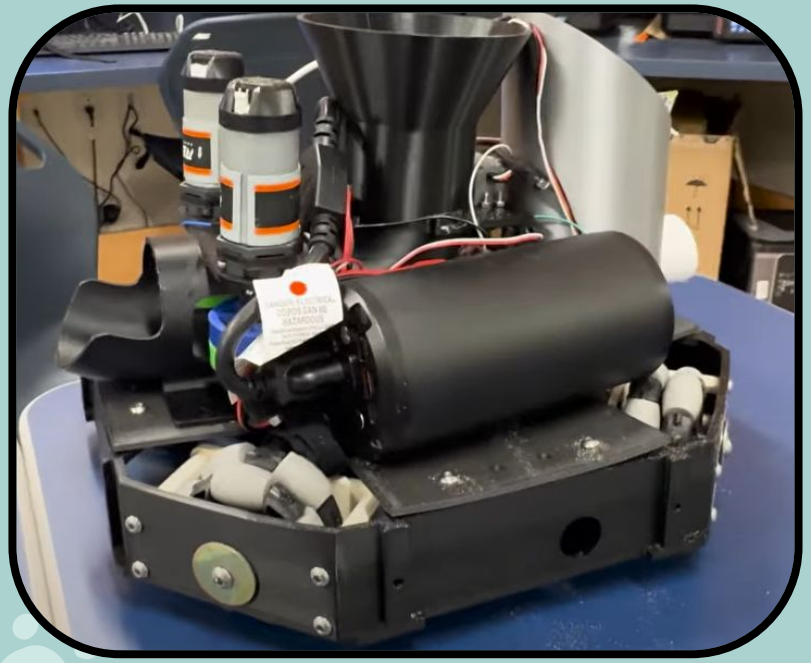
Accomplishments

- Feed pets
- Protect electronics
- Aesthetically pleasing

Planned Features

- Monitor pets
- Feed reliably (stuck kibble)
- Move around or track movement
- Launch tennis ball
- Be easily taken apart/modified

Generation 2



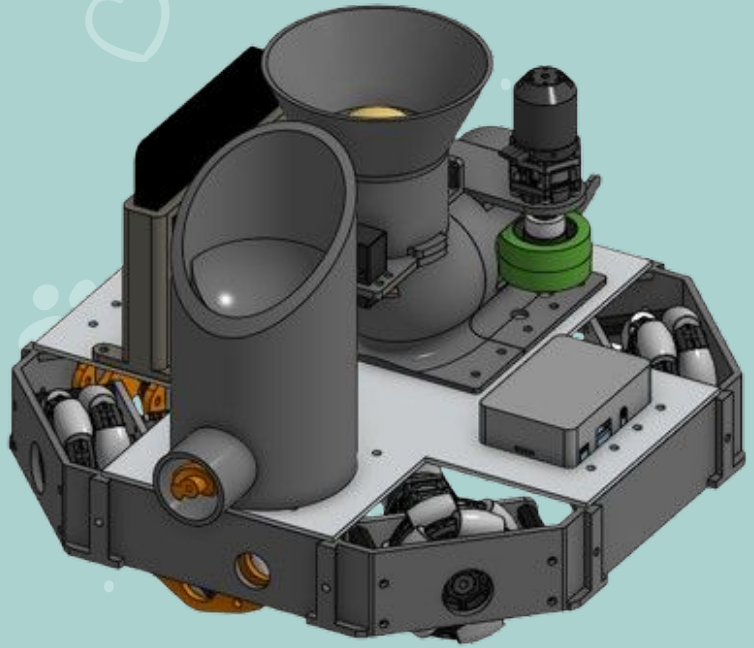
Accomplishments

- Screw Feeder
- Protective Shell (25+ lb)
- Aesthetically pleasing
- Launches Tennis Balls
- Monitors Pets
- Easily modifiable

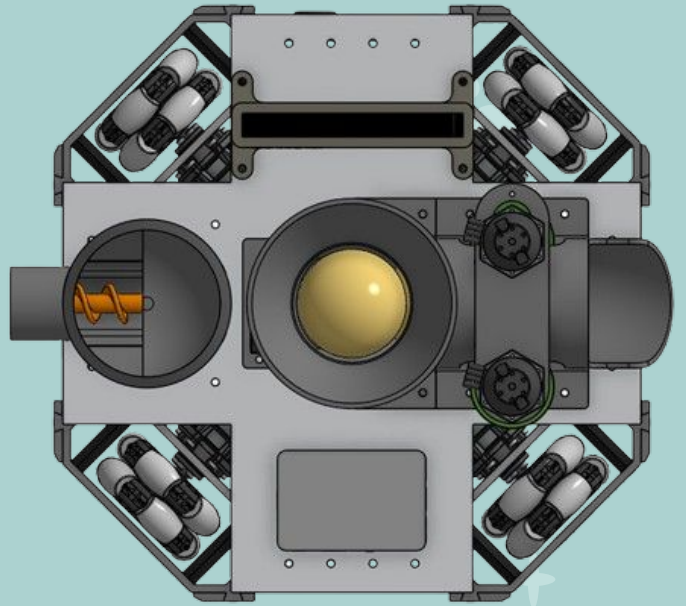
Areas for Improvement

- Can't Move around
- Camera isn’t easily accessible
- Ball Launcher doesn’t launch far
- Not very entertaining for pets

Generation 3 CAD

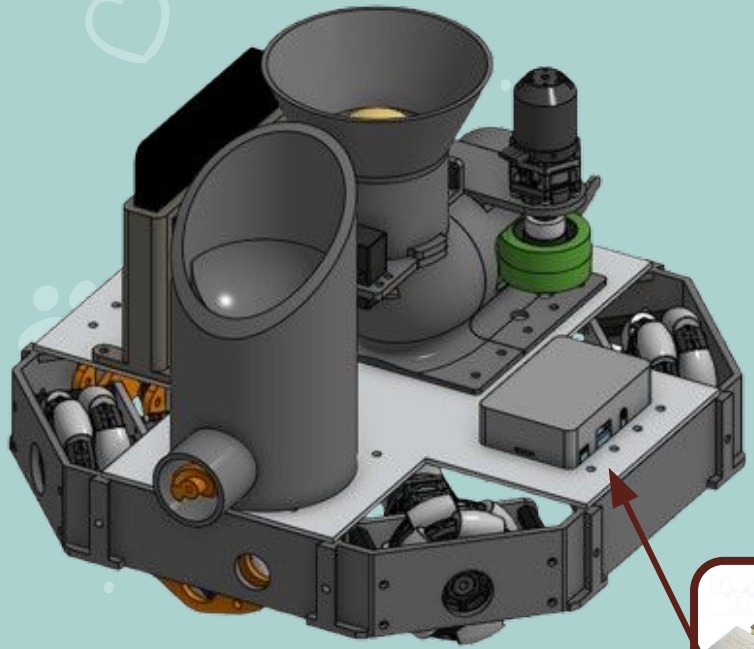


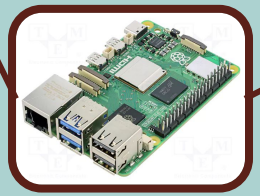
Side View

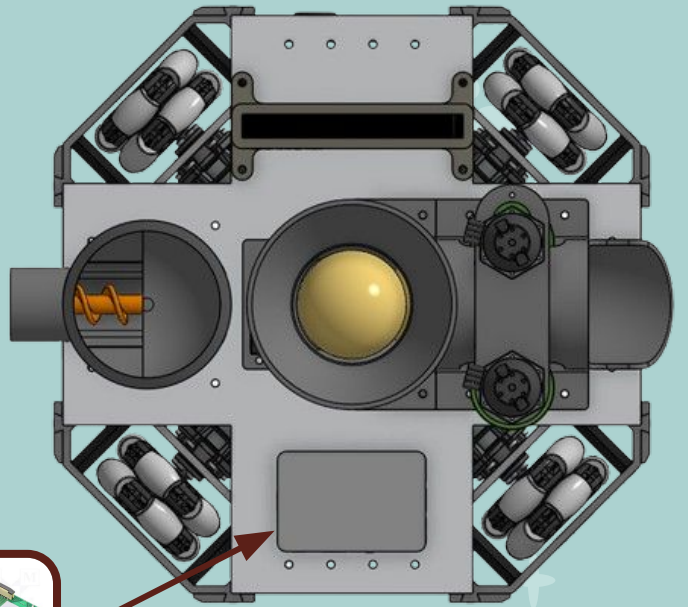


Top View

Generation 3 CAD



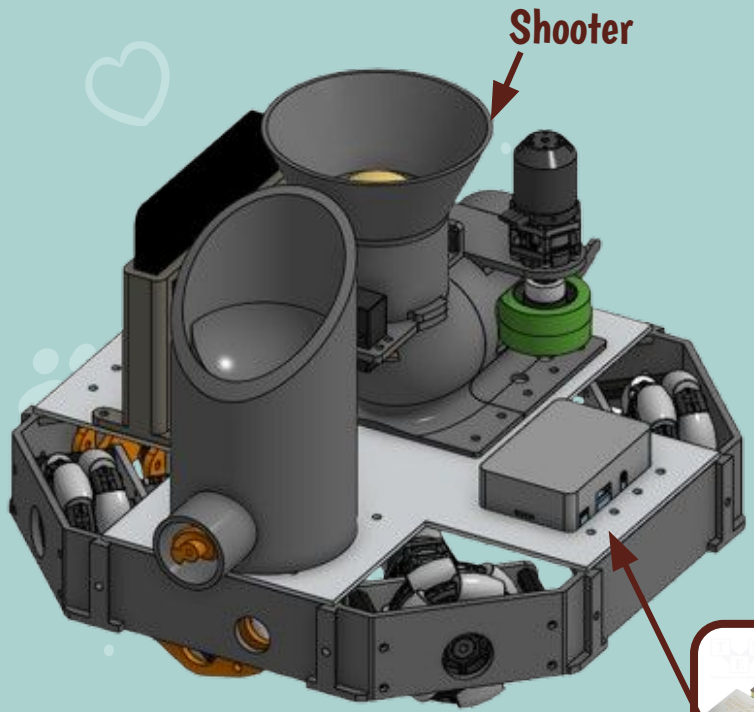




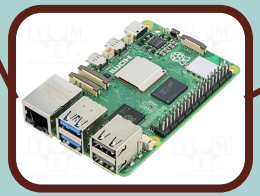
Side View

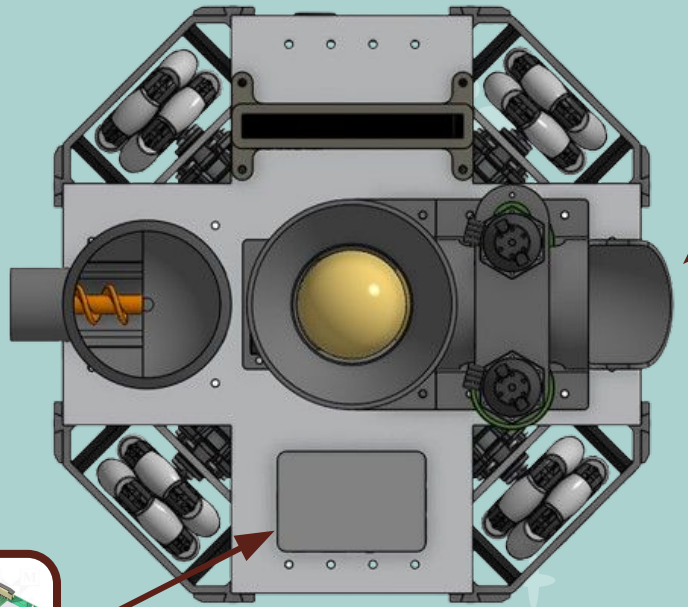
Top View

Generation 3 CAD



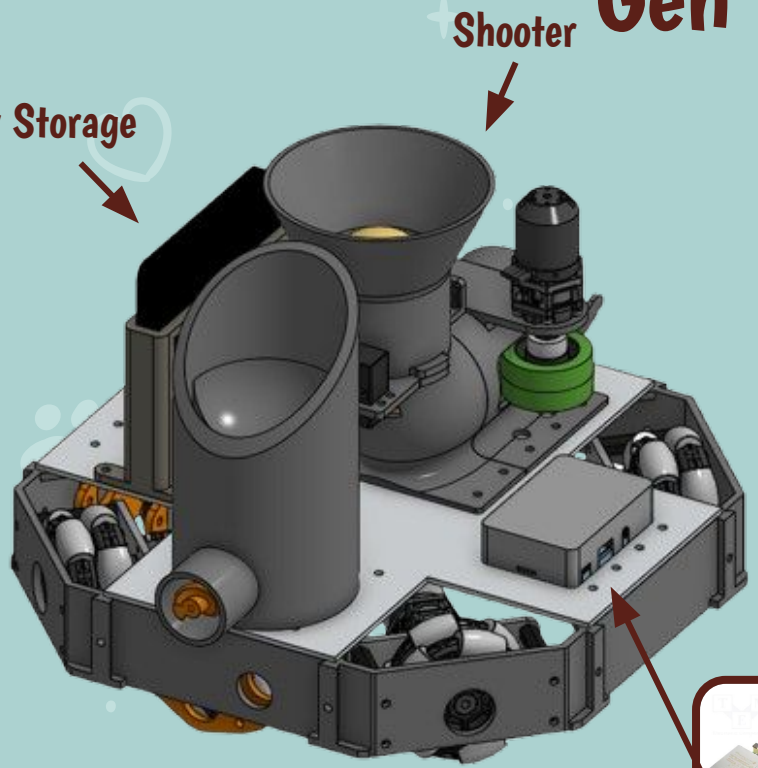
Side View





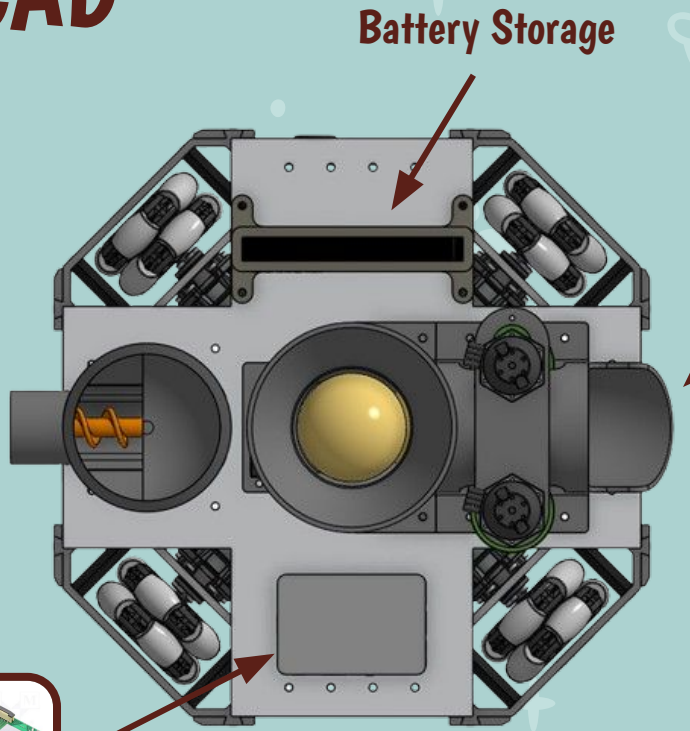
Top View

Gen 3 CAD



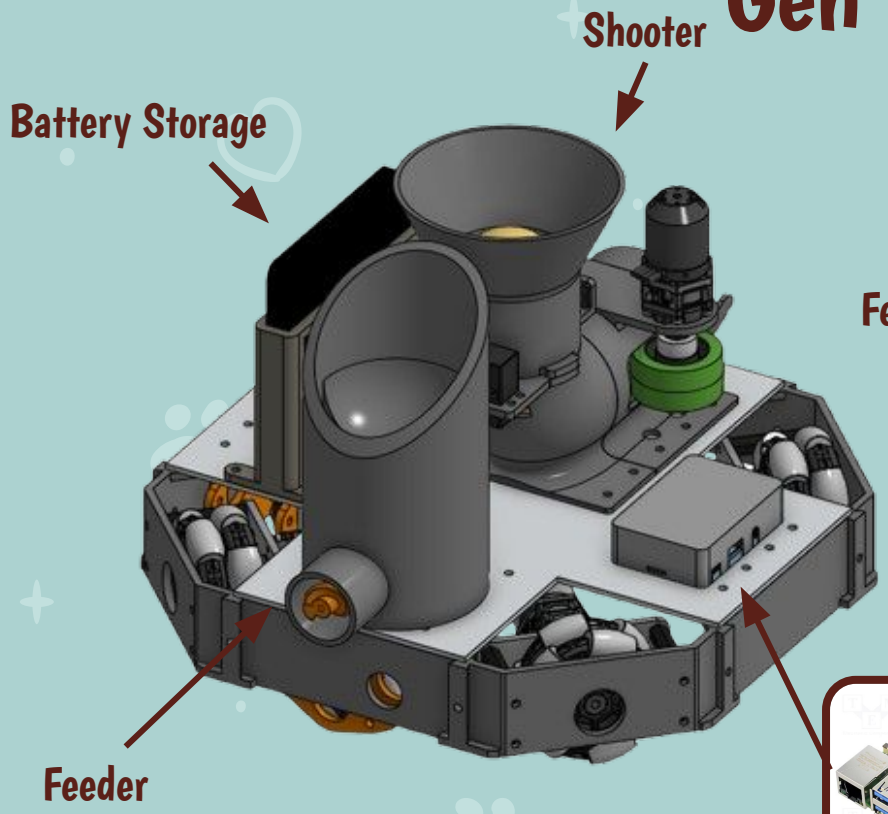
Side View





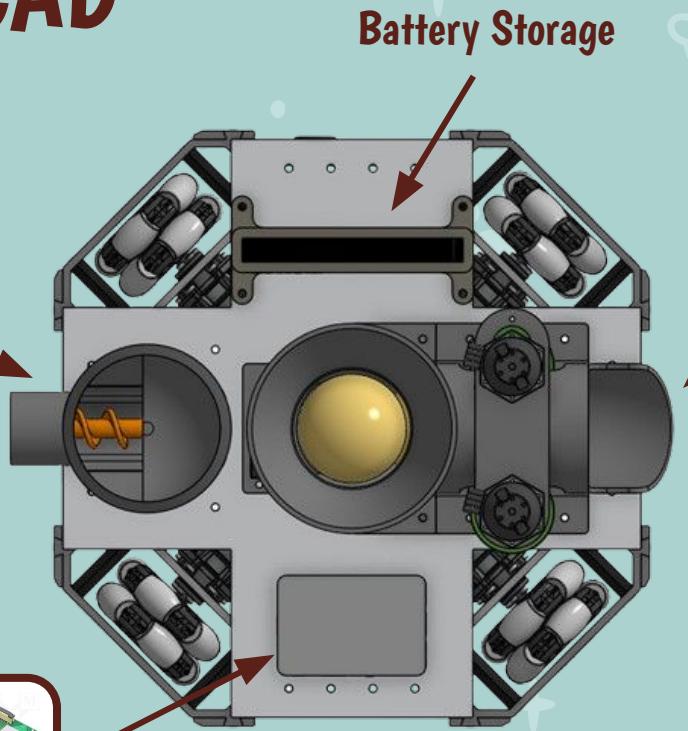
Top View

Gen 3 CAD



Side View





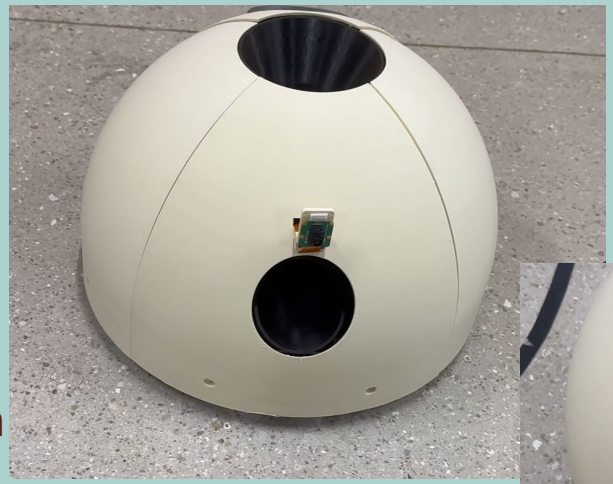
Top View

Generation 3

Features

- Full driving capabilities
- Automatic feeding
- Ball launching with 6’+ distances
- Camera with remote streaming
- Completed dome with high impact resistance
- 27,000 mAh Rechargeable battery

Front





Back

Ball Shooter Test

Analysis

- Mean : 7ft 8in
- Max : 8ft 2in
- Min : 7ft 1in
- Target : 6ft

Trial Number	Distance
1	7ft 1in
2	8ft
3	7ft 5in
4	7ft 6in
5	8ft 1in
6	7ft 3in
7	8ft 2in
8	7ft 9in



Durability Testing

- The top of the dome withstood a downward force of 50 lb (222.4 N)
- No visible deflection, bending, or cracking during or after the test



Video Demo



Bonus: Dog Pictures













